

Gaussian entropy loss under a lower curvature bound

Complete finite-observation proof, sharp constants, and a horizon obstruction

Research working note prepared with AI assistance

7 October 2026

Abstract

For smooth positive densities p, q with $D^2(-\log q) \succeq -\kappa \text{Id}$, finite relative entropy and relative score energy, and an exponential-square moment under p , we prove a weighted upper bound for the entropy lost under common Gaussian convolution. It uses the conditional prediction of the initial relative score. The resulting uniform scalar factor is $(1 - \kappa z)^{-1}$ for $z < 1/\kappa$, and is optimal. Smooth quartic reference densities prove that no finite curvature-only factor exists at or beyond $1/\kappa$, and give an explicit finite counterexample to the unweighted inequality for a nonconvex reference. The argument adapts a public finite-observation lemma; every needed estimate and all limit orders are supplied. It uses no tensor hierarchy or covariance-freezing theorem, and proves no general non-log-concave logarithmic Sobolev inequality. An independent analytic audit supports these statements after correction of one helper-lemma parameter domain. No priority or novelty claim is made.

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1 The precise theorem

For probability densities p, q on $\mathbb{R}^n$, let

$$H(p \mid q) = \int p \log(p/q), \quad w = \nabla \log(p/q), \quad I = \mathbb{E}_p |w(Y)|^2.$$

Let G be standard Gaussian independent of $Y \sim p$, let $U_r = Y + \sqrt{r}G$, and write $p_r = p * N(0, r\text{Id})$, $q_r = q * N(0, r\text{Id})$. Define

$$H_r = H(p_r \mid q_r), \quad J(r) = \mathbb{E}_p |\mathbb{E}_p[w(Y) \mid U_r]|^2 \quad (r > 0).$$

Expectations include the independent observation noise. Set $J(0) = I$.

Theorem 1.1 (Weighted entropy-loss comparison). *Suppose p, q are C^∞ strictly positive probability densities on $\mathbb{R}^n$, $\kappa \geq 0$,*

$$D^2(-\log q) \succeq -\kappa \text{Id}, \quad H_0 < \infty, \quad I < \infty, \quad \mathbb{E}_p e^{c|Y|^2} < \infty \text{ for some } c > 0.$$

For $0 < z < 1/\kappa$, with all $z > 0$ allowed when $\kappa = 0$,

$$2(H_0 - H_z) \leq \int_0^z \frac{J(r)}{(1 - \kappa r)^2} dr, \quad (1)$$

$$2(H_0 - H_z) \leq \frac{1}{1 - \kappa z} \int_0^z J(r) dr. \quad (2)$$

The constants do not depend on dimension. The leading coefficient one in (1) and the scalar factor in (2) are optimal over this class. For each $\kappa > 0$ and $z \geq 1/\kappa$, no finite factor $C(\kappa, z)$ can replace the scalar factor uniformly, already in dimension one.

The quantity $J(r)$ is not generally the relative Fisher information $I(p_r | q_r)$. No identification of these two quantities is used. Data processing gives $0 \leq H_z \leq H_0$. The theorem bounds an entropy difference, not H_0 itself; it does not establish a logarithmic Sobolev inequality for q .

Corollary 1.2 (Conditional version). *Let L take values in a standard Borel space with law λ . Let p_ℓ, q_ℓ and their first derivatives be jointly measurable, with C^∞ strictly positive conditional densities for λ -almost every ℓ . Suppose $D^2(-\log q_\ell) \succeq -\kappa \text{Id}$ with the same κ , and*

$$\mathbb{E}_L H(p_L | q_L) < \infty, \quad \mathbb{E}|w_L(Y)|^2 < \infty, \quad \mathbb{E} e^{c|Y|^2} < \infty$$

for some $c > 0$, where $Y | L = \ell \sim p_\ell$. Take G independent of (L, Y) . Then both bounds hold after averaging the entropies over L and replacing $J(r)$ by $\mathbb{E}|\mathbb{E}[w_L(Y) | U_r, L]|^2$.

Proof. All three finite nonnegative expectations give their one-label finiteness almost everywhere. Apply Theorem 1.1 separately to each such label. Then integrate the proved inequality and use Tonelli on its nonnegative right side. In particular, no uniform mesh-error bound over labels is asserted or needed: the three observation limits are completed at each label before this integration. $\square$

2 Sources and proof dependencies

The source is OpenAI’s *A dimension-free logarithmic Sobolev inequality for subgaussian log-concave measures*, 23 September 2026, pinned at `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Its Lemma 4.2 (pp. 22–25) proves (2) for $\kappa = 0$ with the same entropy, score, and exponential-square assumptions. Its proof was read in full, including every finite-copy comparison and all three limits. Lemma 2.1 and its complete curvature/approximation proof (pp. 6–7) were also read. The following obligations were checked rather than inferred from the main manuscript’s abstract:

1. An adaptive projected observation has the same Gaussian likelihood factor under both priors. The chosen direction is determined by the retained past, so it supplies no additional likelihood ratio.
2. Coinciding projected means give an averaged $O(\delta^2)$ entropy cost, with constants uniform in the finite mesh. Here the posterior curvature is $a = b - \kappa$, not b ; the explicit constant is proved below.
3. Independence is used for full observation arrays across copies. Retained adaptive histories do not factor across copies. Future-dependent residuals are nevertheless conditionally independent given the full past.

4. The comparison of the two quadratic minima uses a dual test and the minimizer of the true minimum. It does not assume that random coordinate projections commute with conditional expectations.
5. Limits are mesh refinement at fixed N, u , then $N \rightarrow \infty$ at fixed u , then $u \rightarrow \infty$. No uniform entropy-cost constant in N is required.

Primary literature was checked at the relevant argument, not only its title. Chen–Eldan [3], Sections 2.4.2 and 3.2.3, gives the Gaussian-quadratic form of localization and the mean-difference entropy drift, Equation (27). That setup uses positive definite continuous-time controls; it does not itself justify our singular adaptive finite observations. Eldan–Koehler–Zeitouni [4], Lemma 2 and its full proof in Section 2.0.1, constructs smooth approximations to function-preserving orthogonal controls for an SDE. Our finite observations require only measurable controls, and no SDE with singular controls is invoked. The positive-curvature criterion is the standard Bakry–Émery input [2]; its precise anisotropic version and extension to finite-energy densities are reproved in Lemma 3.1. The Gaussian channel identity, conditional expectation as an orthogonal projection, entropy chain rule, and the monotone-functions integral inequality are standard.

No theorem about the source’s critical sequence, symmetric tensor hierarchy, matrix square roots, frozen covariance blocks, or scale-integrated information contradiction is imported. In particular, a future use of the source’s main dimension-free LSI would separately require audits of Proposition 5.1, Proposition 6.8, and Propositions 7.2 and 7.9 and their assumptions. They are not dependencies of this note.

3 Positive-curvature and one-step estimates

Lemma 3.1 (Anisotropic curvature inequality). *Let $\pi = Z^{-1}e^{-V}$ be a smooth strictly positive probability on $\mathbb{R}^D$ with $D^2V \succeq S$, where S is a fixed positive definite matrix. For any smooth positive probability density ρ whose right side is finite,*

$$2H(\rho \mid \pi) \leq \mathbb{E}_\rho \langle \nabla \log(\rho/\pi), S^{-1} \nabla \log(\rho/\pi) \rangle. \quad (3)$$

If $S \succeq a\text{Id}$, $a > 0$, and X has law π projected to a k -dimensional subspace and centered there, then

$$\mathbb{E}e^{v \cdot X} \leq e^{|v|^2/(2a)}, \quad \mathbb{E}|X|^2 \leq k/a, \quad \mathbb{E}e^{\theta|X|^2} \leq (1 - 2\theta/a)^{-k/2} \quad (0 \leq \theta < a/2). \quad (4)$$

Proof. The transformation $y = S^{1/2}x$ reduces the first assertion to $S = \text{Id}$. Write $V(y) = |y|^2/2 + U(y)$, where U is convex. First suppose that U is smooth with bounded Hessian. The diffusion $dX_t = \sqrt{2}dB_t - \nabla V(X_t)dt$ has globally Lipschitz drift, is nonexplosive, and preserves π by integration by parts. Its derivative flow satisfies $\dot{A}_t = -D^2V(X_t)A_t$, so $\|A_t\|_{\text{op}} \leq e^{-t}$. For a bounded positive Lipschitz function g bounded away from zero, differentiation of the flow and weighted Cauchy–Schwarz give

$$\frac{|\nabla P_t g|^2}{P_t g} \leq e^{-2t} P_t \left(\frac{|\nabla g|^2}{g} \right).$$

Integration by parts gives $-\frac{d}{dt} \text{Ent}_\pi(P_t g) = \int |\nabla P_t g|^2 / (P_t g) d\pi$. Coupling a trajectory at x with a stationary one using the same Brownian motion bounds their separation by e^{-t} times its initial value. The affine lower bound on U gives Gaussian tails of π ; this coupling therefore yields $P_t g \rightarrow \mathbb{E}_\pi g$. Boundedness and a positive lower bound allow dominated convergence of the entropy to zero. Integration in time gives $\text{Ent}_\pi(g) \leq \frac{1}{2} \int |\nabla g|^2 / g d\pi$. Use $g = f^2 + \varepsilon$ and then $\varepsilon \downarrow 0$ to obtain $\text{Ent}_\pi(f^2) \leq 2 \int |\nabla f|^2 d\pi$ for compact smooth f .

For general convex U , take the Moreau envelopes $U_j(x) = \inf_y \{U(y) + j|x - y|^2/2\}$. They are convex with Lipschitz gradient, increase to U , and a fixed affine minorant $U(y) \geq \ell \cdot y + c_0$ gives

$U_j(x) \geq \ell \cdot x + c_0 - |\ell|^2/(2j)$. Mollify each U_j by a centered compact smooth kernel, choosing its radius so the mollification differs by at most $1/j$ on the ball of radius j . The resulting smooth convex functions have bounded Hessian and the same uniform affine minorant. Their Gaussian-weighted normalized densities converge pointwise and in L^1 , dominated by a fixed integrable Gaussian times a linear exponential. The compact-test inequality passes by dominated convergence.

For $f = \sqrt{\rho/\pi}$ with finite energy, multiply by cutoffs with gradients $O(1/R)$ and approximate locally by compact smooth functions. Their L^2 norms tend to one and their energies tend to the energy of f . Lower semicontinuity of relative entropy follows from $H(\rho \mid \pi) = \sup_h \{\mathbb{E}_\rho h - \log \mathbb{E}_\pi e^h\}$ over bounded h ; normalizing the approximating squared functions thus proves (3). Undoing the linear transformation supplies S^{-1} .

For $S \succeq a\text{Id}$, apply the inequality to exponential tests. If $\psi(t) = \log \mathbb{E}_\pi e^{tv \cdot X}$, it gives $t\psi'(t) - \psi(t) \leq t^2|v|^2/(2a)$. Integrating the derivative of $\psi(t)/t$, also for negative t , proves the first bound in (4). Bounded Lipschitz truncations justify the tests; the Gaussian tails just established allow their passage to the limit. Differentiation at zero gives the covariance bound. Finally average $e^{\sqrt{2\theta}G \cdot X}$ over an independent k -dimensional Gaussian: the first bound yields $\mathbb{E}e^{\theta|X|^2} \leq \mathbb{E}e^{(\theta/a)|G|^2} = (1 - 2\theta/a)^{-k/2}$. This proves all assertions. $\square$

Lemma 3.2 (Uniform entropy cost for matching means). *Fix N , put $D = Nn$, and let the signal $\mathbf{Y} = (Y_1, \dots, Y_N)$ have prior $p^{\otimes N}$. Suppose a retained history $\mathcal{H}$ gives the reference posterior curvature at least $a\text{Id}$, $a > 0$. Let Q be an $\mathcal{H}$ -measurable orthogonal projection for which the two projected posterior means agree. At strength δ , observe $Q(\sqrt{\delta}\mathbf{Y} + G)$. Set*

$$\theta = \min\{c/16, a/4\}, \quad C_{N,a,p} = \frac{4e^{\theta D/(2a)}}{\theta^2} \left[(\mathbb{E}_p e^{8\theta|Y|^2})^N + (1 - 2\theta/a)^{-D} \right].$$

For $0 < \delta \leq \theta$, its conditional relative entropy cost, averaged under the alternative history law, is at most $C_{N,a,p}\delta^2$. The constant is finite and independent of the chosen history sigma-field, mesh, and controls; the bound is averaged, not pointwise in the history.

Proof. Center the projected signal at its common conditional mean and call it X . For the reference, (4) gives $\mathbb{E}_q e^{\theta|X|^2} \leq (1 - 2\theta/a)^{-D/2}$ and $\mathbb{E}_q|X|^2 \leq D/a$. For the alternative let $m = \mathbb{E}_p[\mathbf{Y} \mid \mathcal{H}]$. Since $|Q(\mathbf{Y} - m)|^2 \leq 2|\mathbf{Y}|^2 + 2|m|^2$, two conditional Jensen inequalities give

$$\mathbb{E}_p[e^{\theta|X|^2} \mid \mathcal{H}] \leq (\mathbb{E}_p[e^{2\theta|\mathbf{Y}|^2} \mid \mathcal{H}])^2, \quad \mathbb{E}(\mathbb{E}_p[e^{\theta|X|^2} \mid \mathcal{H}])^2 \leq \mathbb{E}_p e^{8\theta|\mathbf{Y}|^2}.$$

The last expectation is $(\mathbb{E}_p e^{8\theta|Y|^2})^N$, finite since $8\theta \leq c/2$. These bounds hold for every conditioning sigma-field.

In $\text{ran } Q$, let ϕ be standard Gaussian density and h the density of $\sqrt{\delta}X + G$. For a conditionally independent copy X' , direct Gaussian integration gives

$$\chi^2(h \mid \phi) = \mathbb{E}[e^{\delta X \cdot X'} \mid \mathcal{H}] - 1.$$

The linear term vanishes. Use $|e^t - 1 - t| \leq t^2 e^{|t|}/2$, $|X \cdot X'|^2 \leq |X|^2|X'|^2$, and $|X \cdot X'| \leq (|X|^2 + |X'|^2)/2$. For $\delta \leq \theta$, $se^{\delta s/2} \leq (2/\theta)e^{\theta s}$, so

$$\chi^2(h \mid \phi) \leq \frac{2\delta^2}{\theta^2} (\mathbb{E}[e^{\theta|X|^2} \mid \mathcal{H}])^2.$$

Centering and Jensen also give $h_q(x)/\phi(x) \geq e^{-\delta D/(2a)}$ pointwise. Since $H(h_p \mid h_q) \leq \chi^2(h_p \mid h_q)$,

$$H(h_p \mid h_q) \leq 2e^{\delta D/(2a)} (\chi^2(h_p \mid \phi) + \chi^2(h_q \mid \phi)).$$

Average and substitute the moment estimates to obtain the stated constant. For a zero-dimensional observed subspace the cost is zero. $\square$

4 Finite observations and comparison of minima

Proof of (1). Fix z in the stated range, set $b = 1/z$, $a = b - \kappa > 0$, and take N independent copies. Write $\mathbf{W} = (w(Y_1), \dots, w(Y_N))$. All expectations and L^2 inner products in this proof use the alternative $p^{\otimes N}$ together with independent full Gaussian noises. In particular $\|\mathbf{W}\|_2^2 = NI$.

First observe $Z_0 = \sqrt{b}\mathbf{Y} + G_0$. The averaged remaining posterior entropy is $N(H_0 - H_z)$ by the chain rule: the channel is identical under the two priors and its scaled output has law $p_z^{\otimes N}$ or $q_z^{\otimes N}$. Fix $u > 0$, $m \geq 1$, and $\delta = u/m \leq \theta$. Let $\mathcal{H}_i$ be the retained history before step i , initially Z_0 . Given $\mathcal{H}_i$, let P_i project onto the difference of posterior means, with $P_i = 0$ if that difference is zero, and put $Q_i = \text{Id} - P_i$. Retain $Q_i Z_{i+1}$, where $Z_{i+1} = \sqrt{\delta}\mathbf{Y} + G_{i+1}$. Each P_i is measurable in the retained past; positive posterior densities give integral versions of the means. A measurable orthonormal frame in the observed subspace, chosen for example by Gram–Schmidt with a deterministic pivot rule, makes this a usual conditional Gaussian kernel.

For any realized retained history the likelihood as a function of $\mathbf{y}$ is a common Gaussian quadratic factor of precision

$$B = b\text{Id} + \delta \sum_{i=0}^{m-1} Q_i.$$

No likelihood factor is added by choosing P_i , since that choice is a deterministic function of the retained past under both priors. Thus the reference posterior curvature is at least $B - \kappa\text{Id} = a\text{Id} + \delta \sum Q_i$, and its relative score is exactly $\mathbf{W}$. Every reference posterior is proper with Gaussian tails by the affine lower bound on its convex remainder. Alternative posterior means are finite almost everywhere by the exponential-square hypothesis. Lemma 3.2 applies at every step because Q_i kills the posterior-mean difference. The entropy chain rule and Lemma 3.1 for the final posterior imply

$$2N(H_0 - H_z) \leq \mathbb{E}\langle \mathbf{W}, (B - \kappa\text{Id})^{-1} \mathbf{W} \rangle + 2C_{N,a,p}u\delta. \quad (5)$$

All chain-rule entropies are finite in expectation, bounded by NH_0 ; the posterior Fisher right side is at most NI/a .

Work in the Hilbert space of square-integrable D -vector fields on the signal and all full observations through Z_m . Completing the square,

$$aNI - a^2\mathbb{E}\langle \mathbf{W}, (B - \kappa\text{Id})^{-1} \mathbf{W} \rangle = \mathcal{M} := \min_v \left\{ a \|\mathbf{W} - v\|_2^2 + \delta \sum_{i=0}^{m-1} \|Q_i v\|_2^2 \right\}. \quad (6)$$

The minimizer is $v_* = a(B - \kappa\text{Id})^{-1} \mathbf{W}$, so $\|v_*\|_2 \leq \sqrt{NI}$.

Let $\mathcal{F}_i = \sigma(Z_0, \dots, Z_i)$ and let e_i be conditional expectation onto $\mathcal{F}_i$. They are nested orthogonal projections, and P_i is $\mathcal{F}_i$ -measurable. Define the comparison minimum by replacing Q_i in (6) with $\text{Id} - e_i$, and denote it $\mathcal{M}_0$. Its minimizer is

$$v_0 = a \left(a\text{Id} + \delta \sum_{i=0}^{m-1} (\text{Id} - e_i) \right)^{-1} \mathbf{W}.$$

With $\alpha_k = a/(a + k\delta)$, the nested projection decomposition gives

$$v_0 = \alpha_m \mathbf{W} + \sum_{k=0}^{m-1} (\alpha_k - \alpha_{k+1}) e_k \mathbf{W}, \quad (7)$$

$$g_i := (\text{Id} - e_i) v_0 = \alpha_m (\mathbf{W} - e_i \mathbf{W}) + \sum_{k=i+1}^{m-1} (\alpha_k - \alpha_{k+1}) (e_k \mathbf{W} - e_i \mathbf{W}). \quad (8)$$

Indeed a component in $\text{ran}(e_j - e_{j-1})$ has precisely j penalties, hence multiplier α_j ; the final unpredicted component has m penalties. The nonnegative weights in (8) sum to $\alpha_{i+1} \leq 1$.

Conditional on $\mathcal{F}_i$, the full arrays of signal and future noise remain independent across copies: their original joint law is a product, and $\mathcal{F}_i$ conditions each factor only on its own full past. Moreover $e_k \mathbf{W}_j$ uses only copy j 's observations. Thus the blocks $g_{i,j}$ are conditionally independent and centered. The retained histories need not have this property. Conditional Jensen, the tower property, and (8) give

$$Z_{i,j} := \text{tr Cov}(g_{i,j} \mid \mathcal{F}_i) \leq \mathbb{E}[|\mathbf{W}_j - e_i \mathbf{W}_j|^2 \mid \mathcal{F}_i] \leq \mathbb{E}[|\mathbf{W}_j|^2 \mid \mathcal{F}_i].$$

For any integrable $Z \geq 0$ and $M = \mathbb{E}[Z \mid \mathcal{F}]$,

$$\mathbb{E}[M \mathbf{1}_{\{M > K\}}] \leq 2\mathbb{E}[Z \mathbf{1}_{\{Z > K/2\}}],$$

because $M \leq K/2 + \mathbb{E}[Z \mathbf{1}_{\{Z > K/2\}} \mid \mathcal{F}]$. The conditional covariance of g_i is block diagonal, and P_i has rank at most one. Therefore

$$\mathbb{E}|P_i g_i|^2 \leq \mathbb{E} \max_j Z_{i,j} \leq K + 2N \mathbb{E}_p[|w|^2 \mathbf{1}_{\{|w|^2 > K/2\}}].$$

Put

$$\tau_N = \inf_{K > 0} \left\{ K/N + 2\mathbb{E}_p[|w|^2 \mathbf{1}_{\{|w|^2 > K/2\}}] \right\}.$$

Then $\tau_N \rightarrow 0$, by first fixing K to control the integrable tail and then taking $N \rightarrow \infty$. Uniformly in i, m, δ , $\|P_i g_i\|_2^2 \leq N \tau_N$.

No commutation between P_i and e_i is required. For every v , $\|Q_i v\|_2^2 \geq 2\langle v, Q_i g_i \rangle - \|g_i\|_2^2$. The normal equation for v_0 is $a(\mathbf{W} - v_0) = \delta \sum_i g_i$. Consequently the minimum of $a\|\mathbf{W} - v\|_2^2 + 2\delta \sum_i \langle v, g_i \rangle - \delta \sum_i \|g_i\|_2^2$ is attained at v_0 and equals $\mathcal{M}_0$, using $\langle v_0, g_i \rangle = \|g_i\|_2^2$. Test the preceding linear bound at v_* to obtain

$$\mathcal{M} \geq \mathcal{M}_0 - 2\delta \sum_i \|v_*\|_2 \|P_i g_i\|_2 \geq \mathcal{M}_0 - 2uN\sqrt{I\tau_N}. \quad (9)$$

The three limits within the proof

The same orthogonal decomposition gives

$$\frac{\mathcal{M}_0}{N} = \sum_{i=0}^{m-1} \frac{a^2 \delta}{(a + i\delta)(a + (i+1)\delta)} \left(I - \|e_i \mathbf{W}_1\|_2^2 \right).$$

For a component first predicted at step j , the coefficients telescope to $aj\delta/(a + j\delta)$, exactly its minimized quadratic penalty. Full observations through i have sufficient statistic $\sqrt{b}Z_0 + \sqrt{\delta} \sum_{k=1}^i Z_k$. Completing the Gaussian likelihood square shows that all other observation coordinates are independent Gaussian noise with no information about the signal. The resulting precision is $b + i\delta$, not $a + i\delta$. Thus $\|e_i \mathbf{W}_1\|_2^2 = J(1/(b + i\delta))$. Combining (5)–(9) and telescoping yields the explicit prelimit bound

$$\begin{aligned} 2(H_0 - H_z) &\leq \frac{I}{a + u} + \sum_{i=0}^{m-1} \frac{\delta J(1/(b + i\delta))}{(a + i\delta)(a + (i+1)\delta)} \\ &\quad + \frac{2u}{a^2} \sqrt{I\tau_N} + \frac{2C_{N,a,p} u \delta}{N}. \end{aligned} \quad (10)$$

The prediction $J(r)$ is nonincreasing in r and is between zero and I . Indeed a higher-noise Gaussian output is obtained from a lower-noise output by adding independent Gaussian noise; the Markov property and conditional Jensen contract the squared predicted norm. The explicit Gaussian conditional-density formulas also make J measurable. The coefficient $\delta/((a + i\delta)(a +$

$(i+1)\delta))$ is exactly the integral of $(a+t)^{-2}$ over $[i\delta, (i+1)\delta]$. Since $J(1/(b+t))$ is bounded and monotone, this weighted sum converges to its integral as $m \rightarrow \infty$.

First take $m \rightarrow \infty$ at fixed N, u . The explicit $C_{N,a,p}$ is finite at these fixed parameters, so its error vanishes. Next take $N \rightarrow \infty$ at fixed u , using $\tau_N \rightarrow 0$:

$$2(H_0 - H_z) \leq \frac{I}{a+u} + \int_0^u \frac{J(1/(b+t))}{(a+t)^2} dt.$$

Finally take $u \rightarrow \infty$. The remainder tends to zero and the nonnegative integral increases to a finite value bounded by I/a . With $r = 1/(b+t)$ and $a = b - \kappa$, its value is $\int_0^z J(r)/(1 - \kappa r)^2 dr$. This proves (1) in the stated order, without exchanging mesh and copy limits or requiring higher moments of the score. $\square$

Proof of (2). J decreases while $g(r) = (1 - \kappa r)^{-2}$ increases. Integrating $(J(r) - J(s))(g(r) - g(s)) \leq 0$ over $[0, z]^2$ gives

$$\int_0^z J(r)g(r) dr \leq \frac{1}{z} \left(\int_0^z J(r) dr \right) \left(\int_0^z g(r) dr \right).$$

The last integral is $z/(1 - \kappa z)$, including its limit at $\kappa = 0$. Combine with (1). $\square$

5 Sharpness and explicit obstructions

All examples below are one-dimensional smooth positive densities and satisfy the exact hypotheses, including finite entropy and score energy. Fix $t > 0$, $\kappa \geq 0$, and $R > 0$, and set

$$p_t = N(0, t), \quad f_R(x) = e^{\kappa x^2/2 - x^4/R^4}, \quad q_R(x) = Z_R^{-1} f_R(x).$$

Here $0 < Z_R < \infty$ and $(-\log q_R)'' = -\kappa + 12x^2/R^4 \geq -\kappa$. Write $A = \kappa + 1/t$ and $d_R = 4/R^4$. The relative score is $w_R(Y) = -AY + d_R Y^3$, and

$$I_R = A^2 t - 24At^2/R^4 + 240t^3/R^8, \quad (11)$$

$$J_R(r) = \frac{(A - 3d_R t)^2 t^2}{t+r} + \frac{6d_R^2 t^6}{(t+r)^3}. \quad (12)$$

To verify (12), decompose $w_R = (-A + 3d_R t)Y + d_R(Y^3 - 3tY)$ into orthogonal Gaussian Hermite terms. For $U_r \sim N(0, t+r)$, its posterior mean and variance are $tU_r/(t+r)$ and $tr/(t+r)$. Their cubic moment gives the conditional Hermite term $(t/(t+r))^3(U_r^3 - 3(t+r)U_r)$. The two terms are orthogonal, with squared norms $t+r$ and $6(t+r)^3$. This proves (12); $r = 0$ also gives (11).

5.1 Sharp weighted and scalar constants below the horizon

For $z < 1/\kappa$, define $h_R = f_R * N(0, z)$. As $R \rightarrow \infty$,

$$h_R(x) \longrightarrow h_\infty(x) = (1 - \kappa z)^{-1/2} \exp\left(\frac{\kappa x^2}{2(1 - \kappa z)}\right).$$

Completing the square gives the stronger quantitative identity

$$\frac{h_R(x)}{h_\infty(x)} = \mathbb{E}_{T_x} e^{-T_x^4/R^4}, \quad T_x \sim N\left(\frac{x}{1 - \kappa z}, \frac{z}{1 - \kappa z}\right).$$

Jensen therefore yields $0 \leq \log h_\infty(x) - \log h_R(x) \leq \mathbb{E} T_x^4/R^4$. If $X \sim N(0, t+z)$, put

$$T = \frac{t+z}{(1 - \kappa z)^2} + \frac{z}{1 - \kappa z}.$$

Then $0 \leq \mathbb{E} \log h_\infty(X) - \mathbb{E} \log h_R(X) \leq 3T^2/R^4$. No improper prior is used as a probability: h_∞ only evaluates a limit of unnormalized likelihoods, and Z_R cancels from the entropy difference for every finite R .

Since Gaussian differential entropy satisfies $h(p_t) = \frac{1}{2} \log(2\pi e t)$,

$$\begin{aligned} 2(H(p_t | q_R) - H((p_t)_z | (q_R)_z)) &= \log(1 + z/t) + 2\mathbb{E} \log h_R(X) - \kappa t + 6t^2/R^4 \\ &\longrightarrow \mathcal{A}_t(z) := \log(1 + z/t) - \log(1 - \kappa z) + \frac{\kappa z(1 + \kappa t)}{1 - \kappa z}. \end{aligned}$$

Moreover the finite- R expression is at least $\mathcal{A}_t(z) - 6T^2/R^4$. From (12), J_R converges uniformly on $[0, z]$ to $J_\infty(r) = (1 + \kappa t)^2/(t + r)$. Direct differentiation, with both sides zero at $z = 0$, proves

$$\mathcal{A}_t(z) = \int_0^z \frac{(1 + \kappa t)^2}{(t + r)(1 - \kappa r)^2} dr.$$

Thus the ratio in the weighted inequality tends to one for every fixed t, z , establishing optimality of its leading coefficient. For $\kappa > 0$,

$$\int_0^z J_\infty(r) dr = (1 + \kappa t)^2 \log(1 + z/t), \quad \frac{\mathcal{A}_t(z)}{(1 + \kappa t)^2 \log(1 + z/t)} \longrightarrow \frac{1}{1 - \kappa z} \quad (t \rightarrow \infty).$$

Indeed numerator and denominator have leading terms $\kappa^2 t z/(1 - \kappa z)$ and $\kappa^2 t z$. For any smaller proposed factor, first choose finite t large enough, then finite R large enough. This proves optimality with proper smooth probability densities in the order $R \rightarrow \infty$ at fixed t , then $t \rightarrow \infty$. When $\kappa = 0$, numerator and denominator limits both equal $\log(1 + z/t) > 0$, proving optimality of factor one as well.

5.2 A finite counterexample to the unweighted extension

Take $\kappa = t = 1$, $z = 1/2$, $R = 4$. Then $p = N(0, 1)$, $q = Z^{-1} \exp(x^2/2 - x^4/256)$, and q is non-log-concave because its potential has second derivative -1 at zero. Here $T = 7$ and the entropy bound just proved gives

$$2(H_0 - H_{1/2}) \geq \log 3 + 2 - 294/256 = \log 3 + 109/128.$$

The exact prediction integral is

$$\int_0^{1/2} J_R(r) dr = \frac{15625}{4096} \log(3/2) + \frac{5}{12288}.$$

Strict convexity of $x \mapsto 1/x$ gives $\log 3 > 1$ by its midpoint lower bound on $[1, 3]$, and $\log(3/2) < 5/12$ by its trapezoid upper bound on $[1, 3/2]$. Consequently

$$2(H_0 - H_{1/2}) - \int_0^{1/2} J_R(r) dr > \frac{237}{128} - \frac{78145}{49152} = \frac{12863}{49152} > 0.$$

This is an analytic finite counterexample; no numerical quadrature or uncertified estimate of Z is involved.

5.3 No finite factor at or beyond the horizon

Fix $\kappa > 0$, $t > 0$, and $z \geq 1/\kappa$. Using the same proper q_R , symmetry and $\cosh \geq 1$ give

$$\begin{aligned} h_R(x) &= \frac{e^{-x^2/(2z)}}{\sqrt{2\pi z}} \int_{\mathbb{R}} e^{(\kappa-1/z)y^2/2 - y^4/R^4} \cosh(xy/z) dy \\ &\geq \frac{2R}{e\sqrt{2\pi z}} e^{-x^2/(2z)}, \end{aligned}$$

where the last bound integrates just $|y| \leq R$ and uses $\kappa - 1/z \geq 0$. Put $c_z = 2/(e\sqrt{2\pi z})$. The exact entropy-difference identity above now implies

$$2(H_0 - H_z) \geq \log(1 + z/t) + 2\log R + 2\log c_z - (t + z)/z - \kappa t + 6t^2/R^4 \longrightarrow \infty.$$

On the other hand, conditional Jensen and (11) give, for $R \geq 1$,

$$\int_0^z J_R(r) dr \leq zI_R \leq z(A^2t + 240t^3) < \infty,$$

uniformly in R . Every example has finite entropy and Fisher information; the exponential-square moment belongs to the fixed Gaussian p_t . Thus no finite curvature-and-time-only comparison factor exists in this range. This proves the remaining sharpness assertions of Theorem 1.1.

6 Verification and proved boundaries

The accompanying checker uses exact rational arithmetic for Gaussian polynomial predictions, the finite explicit counterexample, precision-kernel identities, and a finite-product test of genuinely adaptive rank-one quadratic-minimum comparisons. Its finite-product noise is discrete: that test probes the Hilbert-space comparison, not the Gaussian entropy-cost lemma. Source hashes and a LaTeX build provide separate provenance and readability checks. They do not substitute for the analytic proofs.

Proved under the displayed hypotheses: the weighted and scalar comparisons, the conditional version, optimal constants, a finite failure of the unweighted nonconvex extension, and impossibility at or beyond the horizon. An independent analytic audit supports the displayed statements. The original Lemma 3.1 allowed negative θ in its square-exponential clause. Its correct domain is $0 \leq \theta < a/2$; all applications already use $\theta = \min\{c/16, a/4\} > 0$. For $\pi = N(0, 1/2)$, $a = 1$, $\theta = -1$, the original bound would assert $1/\sqrt{2} \leq 1/\sqrt{3}$, which is false. This helper-domain correction leaves the main theorem unchanged.

Not established: removal of the exponential-square moment, nonsmooth priors, a general non-log-concave dimension-free LSI, a tensor/freezing extension, priority or novelty, complete Lean verification, or external peer review.

References

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